Farbod Azimmohseni

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Education

M.Sc. **Sharif University of Technology**, Artificial Intelligence — Oct 2025 – July 2027 (expected)
B.Sc. **University of Tehran**, Computer Engineering — Sept 2021 – Sept 2025

- GPA: 19.03/20 • Rank: 1/96

Relevant Coursework

System 2 (Graduate Course)
Sharif University of Technology

- Grade: 19.2/20
- Group 8 project paper

Machine Learning (Graduate Course)
Sharif University of Technology

- Grade: 20/20

Reinforcement Learning (Graduate Course)
Sharif University of Technology

Stochastic Processes (Graduate Course)
Sharif University of Technology

- Grade: 18.2/20

Large Language Models (Graduate Course)
University of Tehran

- Grade: 18.7/20

Research Interests

Reinforcement Learning, Robotics, Continual Skill Discovery, Latent World Models, Compositional Generalization

Publications

Learning Transferable Skills Using Goal-Conditioned Bisimulation 2026
Mohammad Amin Abbasfar, Farbod Azimmohseni, Mohammad Hossein Rohban
Under review at ICLR 2027. Paper

Skeleton-Guided Geometry-Aware Auto-Labeling for Robotic Grasping: From 4-DoF Learning to 6-DoF Execution 2026
Ali Sabzejou*, Farbod Azimmohseni*, Mehdi Tale Masouleh, and Ahmad Kalhor
The International Journal of Robotics Research. Paper

SGG-Net: Skeleton and Graph-Based Neural Network Approaches for Grasping Objects 2025
Alireza Beigy, Farbod Azimmohseni, Ali Sabzejou, Mehdi Tale Masouleh, and Ahmad Kalhor
33rd Iranian Conference on Electrical Engineering (ICEE). Paper

SenseTask: 3D Object Dataset with Task-Specific Annotations for Robotic Grasping 2025
Alireza Beigy*, Farbod Azimmohseni*, Mehdi Tale Masouleh, and Ahmad Kalhor
Preprint, available at SSRN. Paper
*Equal contribution.

Projects

Skeleton-Guided Geometry-Aware Auto-Labeling for Robotic Grasping Paper Link Project Link

- Built an automated straight-skeleton pipeline that generated geometry-aware 4-DoF and 6-DoF grasp labels without manual annotation
- Developed SkelGG-CNN and extended the method to point clouds, achieving 97.4% real-world grasp success on single objects and 95.0% in cluttered scenes

SGG-Net: Skeleton and Graph-Based Neural Network for Grasping Objects Project Link

- Proposed SGG-Net, a hybrid framework combining 3D skeletonization (3D-StSkel) and a Graph Neural Network to generate, evaluate, and rank high-quality 6-DoF grasp poses efficiently
- Developed a fully automated pipeline for 6-DoF grasp generation in cluttered

scenes.

SenseTask: 3D Object Dataset with Task-Specific Annotations for Robotic Grasping Project Link

- Developed a semi-automatic 3D segmentation pipeline for generating comprehensive, task-oriented grasp annotations
- Designed and implemented a user-friendly annotation tool to streamline and accelerate the labeling process
- Curated a dataset of 170 3D models across 14 categories and 4 task types, producing 480,000 grasp annotations for large-scale training

Experience

Robust and Interpretable Machine Learning Lab (RIML), Research Assistant — Sharif University of Technology, Jan 2026 – Present

- Conducting master's thesis research under the supervision of Prof. Mohammad Hossein Rohban, focusing on compositional generalization in reinforcement learning.
- Developing reusable skill representations for zero-shot adaptation to unseen environment dynamics.

Human and Robot Interaction Laboratory (TaarLab), Research Assistant — University of Tehran, Oct 2023 – Jun 2026

- Conducted research under the supervision of Prof. Mehdi Tale Masouleh, developing deep learning methods for robotic grasping and manipulation.
- Contributed to four research projects spanning geometry-aware grasp generation, 6-DoF grasp prediction, and task-specific grasp annotation.

Multi-Agent Reinforcement Learning (Graduate Course), Head Teaching Assistant — Sharif University of Technology
Fall 2026
Instructor: Prof. Mohammad Hossein Rohban

Formal Languages and Automata Theory, Teaching Assistant — University of Tehran
Fall 2023, Spring 2024
Instructor: Prof. Hossein Hojjat

Artificial Intelligence, Teaching Assistant — University of Tehran
Spring 2024, Fall 2025, Spring 2025
Instructors: Prof. Yadollah Yaghoobzadeh and Prof. Hakimeh Fadaei

Data Science, Teaching Assistant — University of Tehran
Spring 2025
Instructors: Prof. Yadollah Yaghoobzadeh and Prof. Behnam Bahrak

Technical Skills

Languages: Python, R, C, C++, Java, Lua

Frameworks: JAX, PyTorch, PyTorch Geometric, scikit-geometry, Transformers, LangChain, LangGraph

Simulation Tools: CoppeliaSim, PyBullet, robosuite, MuJoCo

Honors and Awards

- Awarded Best Bachelor's Project of the Semester by the University of Tehran for work on reinforcement learning in bimanual manipulation
- Graduated as the top student in the Bachelor's program (1/96), University of Tehran.
- Ranked 123rd in the country out of 126,740 participants in the University Entrance Exam of Engineering and Mathematics (Konkur).